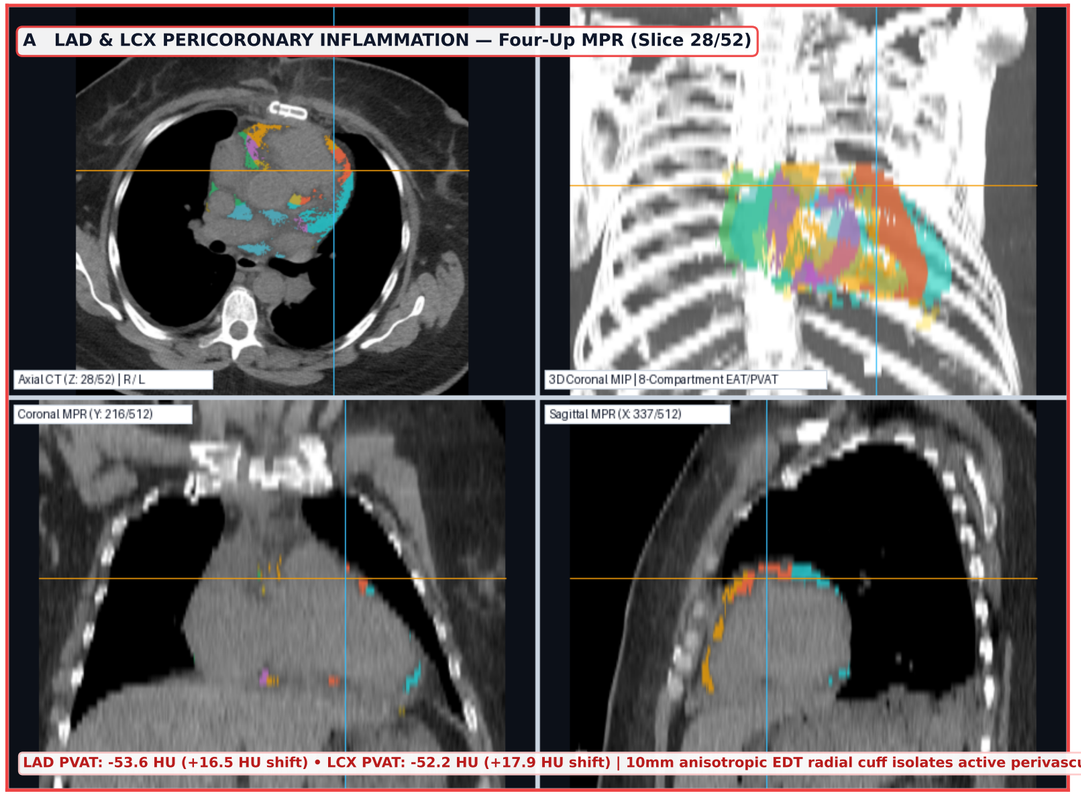
STANFORD COCA CASE STUDY — NONGATED_140: THE ZERO CAC PARADOX & INVISIBLE CORONARY RISK

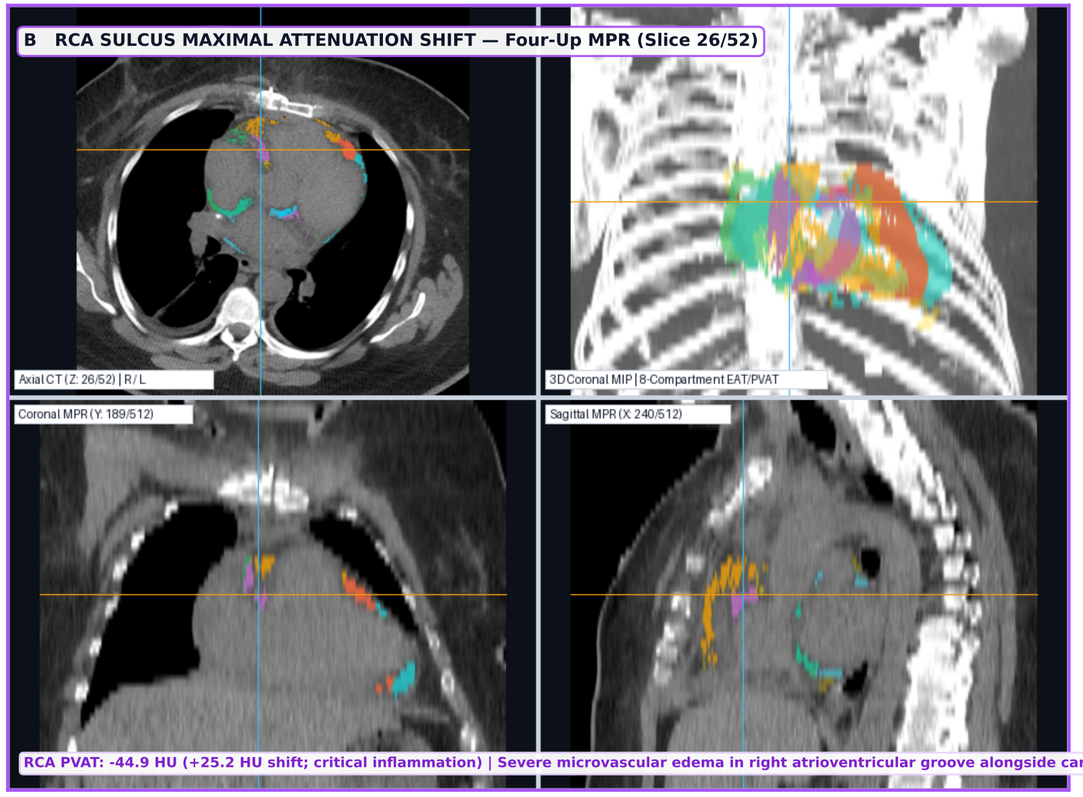
Zero Agatston Calcium Score (CAC = 0.0) Dissociated from High EAT (103.3 cm³) and Multi-Vessel Pericoronary Inflammation (PVAT: -51.1 HU, +19.0 HU Shift)

Agatston CAC: 0.0 (Zero Tier / Missed by Standard Score)

PVAT Total: -51.1 HU (+19.0 HU Shift) • RCA: -44.9 HU (+25.2 HU CRISP-CT Escalation)



LAD PVAT: -53.6 HU (+16.5 HU shift) • LCX PVAT: -52.2 HU (+17.9 HU shift) | 10mm anisotropic EDT radial cuff isolates active perivascular edema



RCA PVAT: -44.9 HU (+25.2 HU shift; critical inflammation) | Severe microvascular edema in right atrioventricular groove alongside cardiomegaly

C 8-COMPARTMENT CARDIAC ADIPOSE & CRISP-CT QUANTIFICATION

Compartment / Biomarker	Volume (cm³)	Mean HU	CRISP-CT Status & Clinical Significance
Agatston Calcium Score (CAC)	**0.0**	**N/A**	**ZERO TIER (Score = 0; falsely classified as Zero Risk)**
Total Epicardial Fat (EAT)	**103.3 cm³**	**-51.5 HU**	**ELEVATED BURDEN (> 100 cm³ cutoff; ~53rd percentile)**
• Left Atrium (EAT LA)	21.9 cm³	-53.0 HU	SEVERE INFLAMMATION (+17.1 HU shift above -70.1 HU)
• Right Atrium (EAT RA)	21.4 cm³	-50.5 HU	SEVERE INFLAMMATION (+19.6 HU shift above -70.1 HU)
• Left Ventricle (EAT LV)	43.2 cm³	-53.6 HU	SEVERE INFLAMMATION (+16.5 HU shift above -70.1 HU)
• Right Ventricle (EAT RV)	40.3 cm³	-49.0 HU	MAXIMAL INFLAMMATION (+21.1 HU shift above -70.1 HU)
Total Pericoronary Fat (PVAT)	**22.5 cm³**	**-51.1 HU**	**DIFFUSE MULTI-VESSEL INFLAMMATION (+19.0 HU shift)**
• LAD Pericoronary (PVAT LAD)	11.2 cm³	-53.6 HU	SEVERE INFLAMMATION (+16.5 HU shift; active biology)
• LCX Pericoronary (PVAT LCX)	5.9 cm³	-52.2 HU	SEVERE INFLAMMATION (+17.9 HU shift; active biology)
• RCA Pericoronary (PVAT RCA)	5.4 cm³	-44.9 HU	CRITICAL INFLAMMATION (+25.2 HU shift; peak edema)

D CLINICAL IMPERATIVE: OVERCOMING THE 'ZERO CAC' DIAGNOSTIC BLIND SPOT

The 'Zero CAC' Blind Spot:	**Agatston scoring detects ONLY calcified plaque (>130 HU). It is 100% blind to non-calcified soft plaque and active**
PVAT Biological Sensor:	**Vascular cytokines (TNF-alpha, IL-6) diffuse outside-in into pericoronary fat, halting adipogenesis and causing ed**
Attenuated Shift (+19.0 HU):	**Edema and vascular proliferation shift PVAT attenuation from quiescent (-80 to -75 HU) to inflamed (-51.1 HU; CR**
AI Radiologist Correlation:	NV-Reason-CT confirms cardiomegaly, postoperative sternotomy changes, and bilateral pleural effusions concordant with high ri

WHY MULTI-PARAMETRIC ADIPOSE PHENOTYPING PREVENTS MISSED CARDIAC EVENTS:

1. Reclassifying 'False-Negative' Low-Risk Patients: Under current clinical guidelines, a patient with CAC = 0 is frequently reassured and denied preventative therapy. Here, monai-cardiac-fat-fl detects critical PVAT inflammation (-51.1 HU) across all 3 vessels.
2. Volumetric + Attenuation Synergy: High EAT burden (103.3 cm³) provides the substrate for metabolic dysfunction, while pericoronary HU shifts capture acute, dynamic vascular wall inflammation months to years before coronary calcification develops.
3. Actionable Prevention: Flags candidates for intensified preventative medical therapy (high-intensity statins, SGLT2i, GLP-1 RA) who would otherwise experience unexpected adverse cardiovascular events despite a pristine calcium score.